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An AI framework for counterattack detection and decision-making evaluation in football

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Abstract

This study proposes a performance analysis framework for evaluating counterattack decisions in football by utilizing deep learning techniques. A dataset of 101,710 frames was selected based on specific algorithmic rules from synchronized StatsBomb event data and tracking data of 580 Premier League matches. Subsequently, a comprehensive approach integrating Transformer and Graph Neural Networks was employed to model and predict match event decisions based on prior match events and tracking data. The results demonstrated that there are approximately 10 counterattacks per match. Each counterattack sequence is typically comprised of 5 events, and 2 of these sequences usually result in shot attempts (successful counterattack). Half of the counterattacks were initiated by defenders, with a success rate of 7.49%. Moreover, no significant monotonic correlation was found between the number of counterattacks per game and the number of goals scored. After modeling and inspecting Permutation Feature Importance, it was shown that player positions, distance advanced, and the relative angle to the carrier are the features with the greatest impact on model performance. Additionally, illustrative examples from the models were provided to aid in the effective analysis of player decision-making during such tactics. The findings of this study can offer strategic insights for coaching and gameplay improvement.

Keywords: Sports analytics, Soccer, Team sports, Deep learning, Decision making, Transformer

Introduction

Football is a low-scoring sport where a single goal (i.e., sparse reward) can substantially impact the outcome of a match. Players constantly make decisions throughout the game to increase their team's chances of winning [1]. Under such highly dynamic context, understanding and optimizing specific decision-making behaviors is crucial for improving team performance. The significance of this process has driven research efforts into analyzing the tactical and technical skills in football, with a focus on how players' decisions influence match outcomes.

The existing investigations have endeavored to measure the impact of various football actions on scoring, such as shooting and movement decisions. The findings have indicated that teams often have certain areas where shooting is more preferable than advancing for a closer attempt, suggesting strategies like increasing long-range shots

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from these areas [2]. Later, researchers compared actual and optimal team decisions to identify areas for improvement in passing and shooting [3]. More recent studies have analyzed offensive and defensive strategies, suggesting that offensive teams focus on long-distance shots from half-spaces, while defensive teams emphasize fouls and ball clearance on the wings and in the center [4]. Additionally, other works have introduced metrics or parameters to evaluate decision-making. For instance, an analysis of the average Decision Value difference between teams' performance in each zone compared to the Premier League average demonstrated its relevance to clubs' transfer activities [5]. Further research has also focused on evaluating passing decisions by balancing risk and reward, highlighting the distinction between players who prioritize safety and those who take calculated risks for greater gains [6]. Similarly, the expected probability of Shot On Target metric offers valuable insights into decisions (probabilities) regarding shooting, blocking, and controlling for each attacker [7]. These methods analyze various in-game scenarios, enabling models to learn optimal strategies and improve performance by simulating decision-making processes under different conditions.

In this extensive exploration of decision-making, counterattacks have emerged as a specific area of interest. Key moments of the game are understood as critical phases during which the balance of play can shift dramatically. These moments are discrete periods that include established attack, transitions (both from attack to defense and from defense to attack), established defense, and set pieces [8]. Each phase of the game presents its own distinct tactical characteristics and opportunities. Transitions, in particular, refer to the dynamic periods during which a team shifts its tactical focus from offense to defense or vice versa [9]. It serves as a crucial tactical in modern football, effectively creating scoring opportunities from a stable defensive position and offering unique strategic advantages, particularly during rapid transitions and ball recoveries [10, 11]. The significance of counterattacks in modern football has recently grown, especially during high-level competitions such as the Premier League and European Championship. Counterattacks constituted 9.5% of attack types in the EPL ($p < 0.001$) and were more effective at creating scoring opportunities compared to combinative attacks (29.6%) and direct attacks (24.1%) [12]. As the Euro 2024 tactical report indicated, key factors such as counter-pressing, maintaining defensive balance, and attacking transitions played a central role throughout the tournament. Transitions alone accounted for 21% of the goals scored ([13] Technical Report, 2024), demonstrating that a team no need to dominate possession to be effective. Instead, the timing and quality of attacking transitions can be equally decisive.

This growing emphasis on transitional play highlights the importance of decision-making once possession is gained. Players must quickly assess whether to launch an immediate counterattack or opt for a more deliberate build-up [14], considering factors such as the opponent's strength, remaining time and current player positioning [15]. However, losing possession in central areas with limited defensive coverage raises counterattack risks, emphasizing the importance of structured build-up play [16]. Most counterattacks originate from the defensive line, last 6–15 s, involving 3–4 attackers against 4–6 defenders, and conclude with a shot, typically between the 46th and 60th min [17]. Key factors such as the player's speed, angle to the goal and the ball, and distance to the goal and the ball have been identified as crucial elements in determining

the success of a counterattack [18]. However, while these studies provide valuable insights into counterattack dynamics, they each have limitations regarding their scope, generalizability, and the complexity of tactical decision-making. Additionally, common methods used to analyze performance in football include action evaluation models such as Expected Threat (xT) Introducing Expected Threat [19] and Valuing Actions by Estimating Probabilities (VAEP) [20]. These models quantify the impact of actions based on event data, with VAEP better capturing the risk-reward tradeoff of actions. In contrast, xT is regarded to be more robust in its assessments due to its basis in positional changes [21]. These models assess each action's contribution (e.g., passes, dribbles) to a team's chances of scoring or preventing a goal, accurately evaluating player actions and their impact on goal outcomes. However, all these models assess the value of the actions taken by the player and coach, yet they fail to offer recommendations for alternative or optimal actions. Consequently, there is a gap in the analysis of decision-making within the specific tactical context of counterattacks in football.

Predicting counterattack decisions in soccer presents several challenges due to the game's inherent complexity. A match spans a long duration, involves numerous players and features an ever-changing game environment. This highly dynamic nature makes it arduous to measure and evaluate decisions, as effective decision-making during counterattacks requires players to rapidly assimilate information from various sources, including the positions of teammates and opponents, spatial awareness, and the evolving state of play. Modeling these processes is further complicated by the unstructured nature of the data [22]. Accurately representing the state of the game is central to football analytics, necessitating a paradigm capable of handling sequential decision-making, sparse rewards, and intricate player-ball interactions [23]. Furthermore, the success of a counterattack depends not only on immediate outcomes but also on the cumulative influence of numerous seemingly minor actions throughout the match [24]. Assessing whether a particular decision was optimal is inherently challenging, if not impossible, due to the multitude of interacting variables that contribute to the outcome of each action. Additionally, accurately identifying the sequences that lead to counterattacks poses a considerable challenge. Existing studies often rely on manual annotation [25], which is both time-consuming and labor-intensive. Subsequent attempts have involved using rule-based algorithms [16] as well as video detection techniques to automatically identify transition phases [18, 26]. However, their accuracy largely depends on the data structure and the validity of the algorithmic rules employed.

To address these challenges, recent studies have applied deep learning techniques that can handle the game's inherent complexity and dynamism. Convolutional Neural Networks (CNNs) are adept at capturing spatial relationships, enabling the analysis of player and ball positions in the dynamic game environment. Similarly, Recurrent Neural Networks (RNNs), including Long Short-Term Memory (LSTM) networks [27] and Gated Recurrent Units (GRU) [28], are effective in modeling temporal sequences, which is crucial for understanding the sequential decision-making processes during counterattacks [27, 29–31]. Transformer encoders have been used to convert historical event vectors into new representations, effectively modeling temporal dependencies and dynamic changes during counterattacks. This facilitates rapid information assimilation, akin to the cognitive demands on players during a match [32, 33]. Furthermore,

deep learning methods can automate the identification of sequences that lead to counterattacks, reducing dependence on manual annotation and the constraints of rule-based algorithms. By learning patterns directly from the data, these models enhance the accuracy of counterattack detection. Despite these advances, current approaches still struggle to effectively model the players' relationships and sequential dependencies, which are vital for accurate decision-making in football. The main challenge lies in integrating diverse data types, such as player positions and match events, into a unified model that can inform tactical planning.

Accordingly, the study aims to identify counterattack sequences and develop a comprehensive model that predicts optimal decision-making outcomes during counterattacks when a team gains possession can provide valid insights into team performance in elite football. It simulates the actual context in which players make decisions and the factors they consider, while also deepening the understanding of influences on these critical choices. By conducting a predictive analysis of player decision-making, the results can be evaluated to improve athletic performance, leading to more effective counterattacks and overall better team performance. It offers significant insights into decision-making processes in football counterattacks, which coaches and analysts can use to enhance tactical planning and training methods.

Methods

Sample and data

The dataset used in this study was provided by StatsBomb and contains 946,280 events from paired event and 360 data, covering 580 matches from the Premier League during the 2022/23 and 2023/24 seasons. The event data describes on-the-ball actions, such as passes, dribbles, and shots, along with their time of occurrence, origin and destination locations, the player executing the action, the action's outcome, and the body part used. The SPADL (Soccer Player Action Description Language) representation of the event was employed and modified for the subsequent analysis [20], adding ball recovery actions (Supplementary Table 1). The 360 snapshots are captured at the moment of each on-the-ball action and contain the locations of all visible players as well as their relationship to the ball carrier (i.e., teammate or opponent) [34].

Data preprocessing and feature engineering

A possession sequence is defined as a series of consecutive actions performed by one team without interruption by the opposing team [35]. In this study, sequences originating from dead-ball situations, except for throw-ins (e.g., corners, free-kicks), were excluded. Additionally, sequences with fewer than four players in the 360 snapshot or those containing fewer than three events were omitted to ensure data quality and consistency. The event data utilized in the Transformer model includes a range of features, such as game time, the ball carrier's tactical position, event type, duration, event outcomes, start and end coordinates, delta x and delta y, distance change to the goal, angle changes between events, VAEP, On-Ball Value (OBV), play patterns, and game momentum, Features that exhibited both positive and negative values were scaled using a MinMaxScaler with a range of (− 1, 1), while features containing only positive values were scaled within the range of (0, 1). For the GATs

model, the features include player coordinates, player role information, distance to the ball, angle to the ball, ball carrier pressure status, live goal difference, and the centroid of their respective team cluster (as shown in Fig. 1).

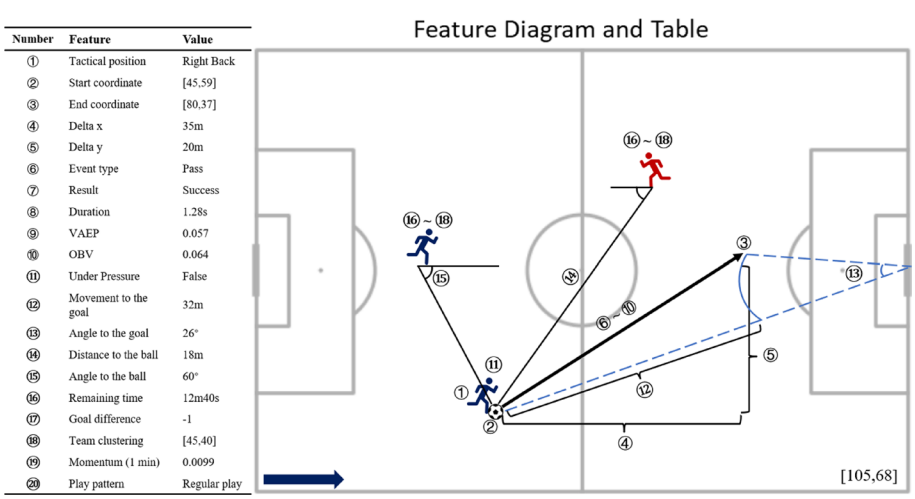
Subsequently, algorithmic rules based on Statsbomb's definition [36] and relevant literature[18] were applied to determine whether an event is part of a counterattack sequence. It assigns labels to each instance in the dataset by analyzing the sequence of on-ball event data to identify either a build-up or a counterattack. To detect counterattacks, the following rules are applied:

i. The sequence started with an open play turnover inside the counter-attacking team's defensive half of the pitch.
ii. The sequence of events does not contain freekick and corner.
iii. The sequence was at least 75% directed toward the goal.
iv. The ball moves at least 16 m forward.
v. The duration of a counterattack is at most 14 s.

Afterwards, two football performance analysts with more than 3 years of collecting and analyzing football data, annotated counterattack sequences using video footage from three randomly selected matches to verify the reliability of algorithm. The Kappa consistency test (conducted in SPSS) indicated that the algorithm's labeling results were effective, with an average Kappa coefficient higher than 0.9. Finally, a total of 5763 counterattacks, containing 101,710 actions (including build-up plays) were extracted from the sampled matches.

Model construction

In the study, a fusion model integrating Graph Attention Networks (GATs) [37] and Transformer [38] was proposed for predicting counterattack decisions, as illustrated



Number	Feature	Value
①	Tactical position	Right Back
②	Start coordinate	[45,59]
③	End coordinate	[80,37]
④	Delta x	35m
⑤	Delta y	20m
⑥	Event type	Pass
⑦	Result	Success
⑧	Duration	1.28s
⑨	VAEP	0.057
⑩	OBV	0.064
⑪	Under Pressure	False
⑫	Movement to the goal	32m
⑬	Angle to the goal	26°
⑭	Distance to the ball	18m
⑮	Angle to the ball	60°
⑯	Remaining time	12m40s
⑰	Goal difference	-1
⑱	Team clustering	[45,40]
⑲	Momentum (1 min)	0.0099
⑳	Play pattern	Regular play

Fig. 1 Graphic illustration of features extracted from counterattack. The serial number denotes the name and value (example) of each indicator, accompanied by a diagram that elucidates its definition. The individuals depicted in blue represent teammates, while those in red indicate opponents. The arrow positioned in the lower left corner indicates the direction of attack, while the coordinates displayed in the lower right corner denote FIFA-standard football field dimensions (105 m × 68 m)

in Fig. 2. The primary objective of this model is to assess decision-making during key moments of possession, focusing on determining whether a player should initiate a quick counterattack or retain possession. The model was trained on a dataset where 80% of the sequences (determined by sequence proportion) were used for training, with a balanced distribution comprising 55% build-up actions and 45% counterattack actions. A randomly selected 10% of the dataset was used as a validation set for model selection.

The proposed model consists of two main components. First, the GATs, which have demonstrated robust performance in graph-based tasks, are employed to capture relational data between players and the underlying structure of the team's network. Second, the Transformer model was used to model the temporal sequences of play, enabling it to account for the dynamic flow of events over time. The outputs of both models are concatenated and passed through a final classification layer, which predicts the optimal strategy for the player under the given circumstances.

Subsequently, the predicted value of a player's action is utilized to conduct a comprehensive analysis of the decision-making process. This analysis is further extended using game theory to optimize interaction strategies, enhancing the team's overall performance.

GAT component

The GAT component was designed to capture the spatial interactions and relationships among players on the field. It utilizes the GATConv layer, which performs multi-head attention on graph-structured data. The model includes two GAT convolutional layers. The first layer processes the player feature matrix x and the edge information, which encodes player interactions, applying multi-head attention. This generates a hidden representation, which is subsequently refined by a second GAT layer. Dropout regularization and ELU activation functions are applied between layers to prevent

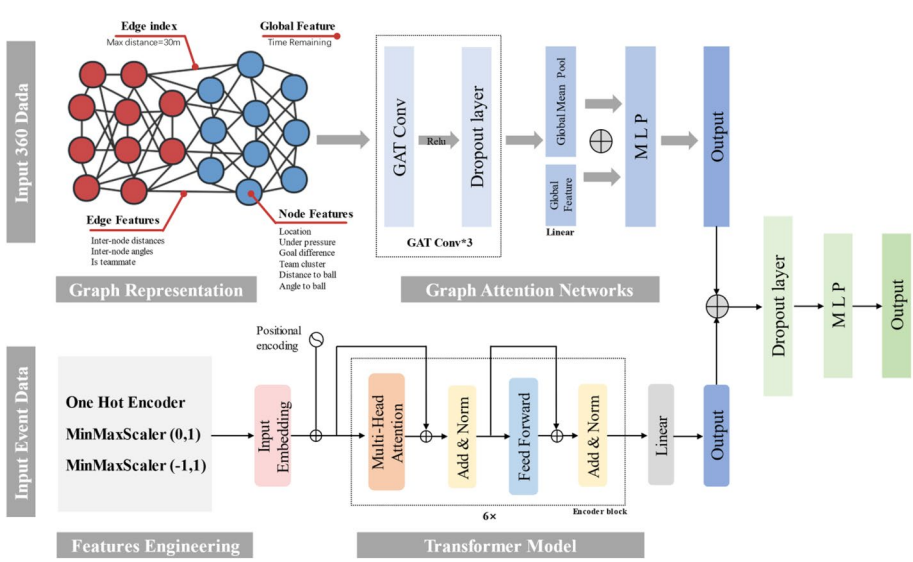


Fig. 2 The predictive model architecture. the upper section illustrates the process of Graph Attention Networks (GATs), while the lower section employs the Transformer model

overfitting and introduce non-linearity. The final output represents a compressed feature space that captures the player interactions and team coordination.

Each frame of the 360 data is transformed into a graph representation to train the GNN [39] using PyTorch. In this graph, each node represents a player, and node features include player coordinates, player role information, distance to the ball, angle to the ball, whether the ball carrier is under pressure, live goal difference, and the centroid of their respective team cluster. Node features are summarized in Table 1.

Edges between nodes are established based on the Euclidean distance between players, which helps the GNN understand spatial relationships. If the distance between two players is less than a threshold of 30 m, an edge is created between them. Each edge carries attributes such as the distance between players, their relative angle, and team affiliation.

Transformer component

Each sequence, consisting of three consecutive events, is fed into a Transformer model to capture temporal dependencies. The Transformer employs a multi-head self-attention mechanism to highlight pivotal moments in the game by computing attention scores that weigh the importance of each event within the sequence. Positional encoding is applied to preserve the order of events, ensuring that temporal relationships are maintained. The model's output represents key features from the event sequence, contributing to downstream decision-making tasks.

The outputs from both the GATs and Transformer models are concatenated along the feature dimension and passed through a multi-layer perceptron (MLP). This layer reduces the combined feature vector to two logits, representing the choice between initiating a counterattack or opting for a build-up play. By integrating both spatial information (via GATs) and temporal information (via Transformer), this fusion architecture provides a comprehensive understanding of player decision-making in critical match situations.

Model calibration and feature importance

Model calibration: In classification tasks, machine learning models typically generate probabilistic outputs within a range of 0 to 1, which are then utilized to inform decision-making processes. However, these probabilistic outputs do not always reflect the true likelihood of an event occurring. This discrepancy arises because many machine

Table 1 Node features for graph attention networks

Node features	Description
Location	Player coordinate [X, Y]
Under Pressure	Indicates whether the player with the ball is under pressure
Goal Difference	Team live score difference
Team Cluster	The centroid of each team
Relative position to the ball	Distance to the ball and angle to the ball
Time Remaining	The real-time remaining time of the game, which applies to all nodes

learning models, while effective at classifying, do not naturally produce well-calibrated probability estimates. To ensure that the model's output reliably estimates the actual probability of an event, calibration is necessary.

This can be measured using the Expected Calibration Error (ECE), which calculates a weighted average of the difference between absolute accuracy and confidence for each probability bin.

Feature importance: To gain deeper insights into the decision-making process of the proposed model and to identify the most influential features in its predictions, Permutation Feature Importance (PFI) [40] was employed by performing a random shuffle on each feature 20 times. This method quantifies feature importance by assessing the reduction in model performance when a feature's values are randomly permuted, thereby disrupting its association with the target variable [41].

In the implementation of PFI, all features in the model were systematically iterated, including player-specific attributes (e.g., coordinates, pressure), global game contexts (e.g., time remaining), and event-based features (e.g., play patterns, event types, and tactical positions), the edge features are excluded from the scope of this feature importance analysis, as shuffling them would break the inherent structure of the graph and lead to misleading conclusions. For each feature, multiple permutations were performed by randomly shuffling its values across the test set, while leaving all other features unchanged. The model's accuracy on the permuted dataset was then measured and compared to the baseline accuracy. The average reduction in accuracy across these permutations provided a robust measure of feature importance.

Model settings

The GAT model's input feature vector dimension was 64, with 4 attention heads. The Transformer had an embedding dimension of 256 and a hidden layer size of 256, using 8 attention heads. Cross Entropy Loss (CEL) was used to measure prediction error. The model was trained with the Adam optimizer, using a learning rate of 0.001, and a batch size of 256, for 200 epochs. The checkpoint with the lowest validation loss was chosen as the final model (Supplement Table 3).

The baseline model

To assess the performance of the proposed model, a comparison was conducted with XGBoost and Random Forest models. These machine learning models were trained using the same event data used in the proposed model, ensuring consistency in data representation and feature engineering. Both models were optimized using grid search to determine the best hyperparameters for each.

Results

The proposed model achieved a precision of 83.55%, a ROC-AUC score of 0.9132, and a log loss of 0.3559 (see Fig. 3a). The higher precision and ROC-AUC scores, combined with the lower log loss, showed a clear improvement in performance over the baseline models. The calibration of the proposed model, assessed using ECE, showed a value of 0.017, indicating a strong alignment between predicted probabilities and actual outcomes. Calibration curves in Fig. 3 reveal minimal discrepancy between predictions

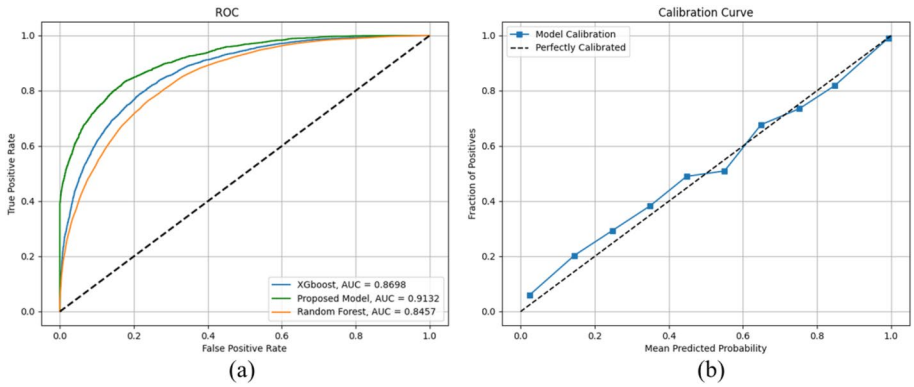


Fig. 3 Performance and calibration of the proposed model. The proposed and baseline model ROC (**a**) and calibration curve (**b**)

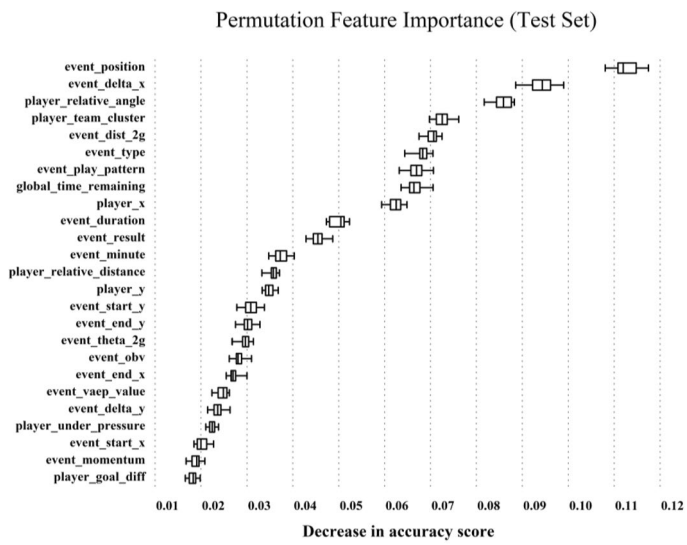


Fig. 4 The proposed model's feature importance (test set). Permutation feature importance is ranked by the impact on model accuracy; "player" features represent GATs, and "event" features represent the transformer model

and observations, suggesting the model excels in classification and provides well-calibrated probabilities for practical decision-making. (see Fig. 3b).

Feature importance

Figure 4 presents the permutation feature importance of the proposed model. Among all features, player positions, delta x, and relative angle to the actor showed the most significant impact on model accuracy, as indicated by their higher mean importance scores (0.08–0.12) and lower variability. Contextual features such as event type (0.07) and remaining game time (0.07) also demonstrated notable importance. In contrast, features like momentum and goal difference had the lowest importance scores, whereas event play patterns, exhibit greater variability in their importance, suggesting inconsistent contributions to the model's accuracy.

Counterattack performance evaluation

After applying the current model to automatically detect counterattacks in Premier League matches, it was shown that there was an average of 10 counterattacks per match. Each sequence typically involves 5 events, and 2 of these sequences lead to shot attempts. No significant monotonic correlation was found between the average number of counterattacks per game and goals scored during the 2022/2023 and 2023/2024 Premier League seasons (Spearman's $\rho = 0.0233$, $p = 0.5794$). Figure 5 shows that Manchester City (end-of-season ranking: 1) and Arsenal (end-of-season ranking: 2) in 23/24 showed a higher conversion rate (25.8%). Liverpool of 23/24 season (end-of-season ranking: 3) had the highest average counterattacks per match, with a decline in efficiency compared to the previous season (end-of-season ranking: 5). Newcastle showed a decrease in both counterattack times and efficiency in 23/24 season (end-of-season ranking: 7). Although Manchester United showed a slight increase in counterattacks in 23/24 (end-of-season ranking: 8), their shots declined.

Tactical roles in counterattack

In this study, a successful counterattack is defined as one that results in a shot taken within the opponent's penalty area. Figure 6 shows the role of different player positions (Supplement Table 2) in initiating counterattacks and their likelihood of leading to shots. Defenders (positions 2–8), particularly positions 3 and 5, showed higher numbers of counterattack initiation than the rest of the teammates, while position 7 had the highest success rate of 10.16%, Forwards (positions 17–24) recorded the least initiated counterattacks but success rates were relatively consistent around 6–8%.

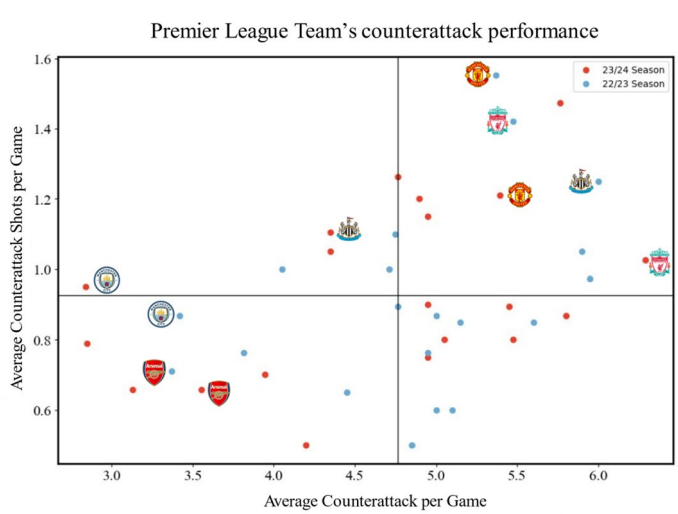


Fig. 5 Average counterattacks vs. shots per game (22/23 and 23/24 seasons). It compares average counterattacks per game with average shots per game for selected teams across the 22/23 (blue) and 23/24 (red) seasons

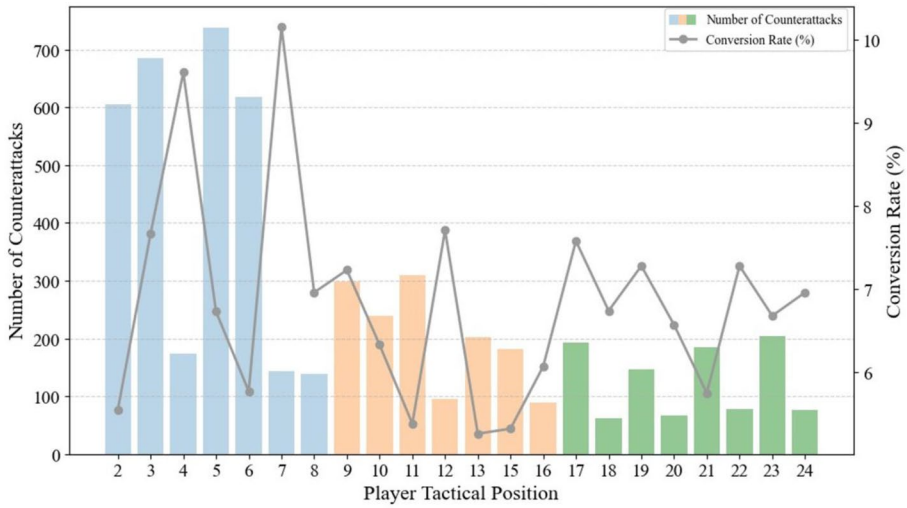


Fig. 6 Frequency of counterattack initiation and conversion rate for different tactical positions. Blue bars denote defenders, orange for midfielders, and green bars for forwards; gray line represents the conversion rate across different tactical positions

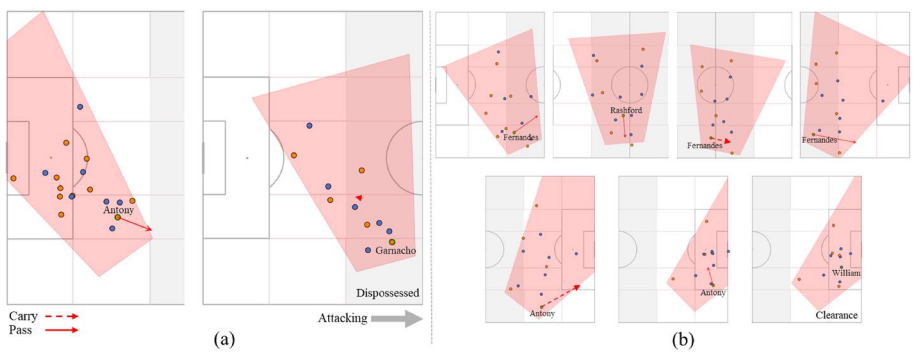


Fig. 7 Two counterattack scenarios within the match between Manchester United vs Wolves. The image (**a**) shows the counterattack at 26m34s, while the image (**b**) shows another at 41 m 57 s. The match was on Aug 14, 2023

Model application

Figure 7 exemplifies the visual comparison between the model's predictions with players' actual in-game decisions. Figure 7a illustrates the specific game scenario in which the model predicted a build-up after Manchester United regained the ball possession, but Antony's carry was tackled, providing Wolves with an opportunity to attack after the gegenpressing. Another scenario in Fig. 7b shows the model predicted Manchester United's actual counterattack decision during the whole sequence exceeding 14 s.

The study further compares outcomes when teams adhere to their behavioral decisions versus optimal decisions. Figure 8 illustrates ball recoveries made by two teams and the quality of chances generated from those situations. While Manchester United managed to regain more possessions from Liverpool during their opponent attacks, they created less dangerous opportunities from these recoveries (see Fig. 8a). Moreover, Liverpool had slightly more successful regains occurring in their left wing

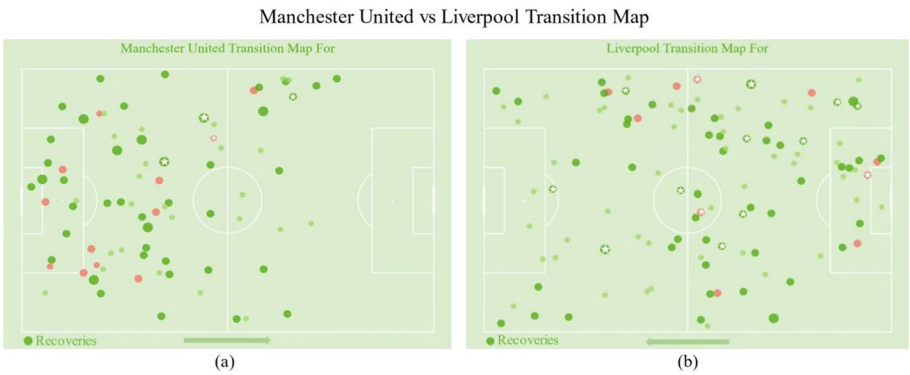


Fig. 8 Manchester United vs Liverpool on Apr 7, 2024. The size of each circle represents the quality of the opportunity, and a star indicates a shot attempt. The larger green circles represent situations where both actual and optimal decisions are to counterattack. Smaller green circles indicate that both decisions favor a build-up. Red-colored circles showcase where behavioral and optimal decisions differ; larger circles represent the decision was a build-up while the optimal was a counterattack, and smaller circles represent that the decision was a counterattack while the optimal was a build-up

of Manchester United's half, compared to its right wing. They effectively convert regains in their own half into successful counterattacks, (see Fig. 8b).

Discussion

This study presents a novel evaluation framework for possession decisions in football, focusing specifically on counterattacks. Meanwhile, player positions, the distance advanced, and the relative angle to the actor were shown to play the most critical role in determining counterattack decisions. An important contribution of this work is demonstrating how decomposing a complex decision-making process into smaller components restores the actual context of players' decision-making. This approach enhances interpretability, enabling more targeted analysis and providing coaches and analysts with reliable predictions of players' actions during counterattacks.

The results reveal that player positions, the distance advanced, and the relative angle to the actor are key factors influencing counterattack decisions. It is often assumed that increasing the number of counterattacks inherently leads to higher goal-scoring efficiency. However, our study challenges this belief, suggesting instead that factors such as the counterattack quality, timing, and situational context of counterattacks play a more critical role [42, 43]. Specifically, understanding tactical roles is crucial for players, as it enables them to maximize their effectiveness on the field by enhancing their positional skills and decision-making abilities. Consequently, this proficiency allows them to create opportunities for both themselves and their teammates, thereby significantly contributing to the success of the team's counterattacks [17].

Effective counterattacks depend not only on the players' tactical position but also on their ability to make quick, informed decisions under pressure. For example, wide players must possess a keen sense of tactical awareness to exploit available spaces, utilizing both their speed and spatial intelligence to disrupt the opponent's defensive structure [44]. The ability to be adept at finding space in dangerous areas or creating these spaces can gain better counterattack depth and width [43]. Midfielders play a key role in enabling quick transitions from defense to attack, acting as the link between the two phases. Their

ability to quickly regain possession and execute incisive passes or progress with the ball under pressure is crucial for capitalizing on fleeting opportunities. The balance between their attacking and defensive responsibilities is pivotal in ensuring that the team can capitalize on such moments [45]. Center-backs have increasingly become integral to the initiation of counterattacks, particularly in possession-based systems. Their ability to deliver precise long passes or to dribble directly from the back often triggers rapid transitions from defense to attack. As key decision-makers in defensive situations, center-backs must exhibit both vision and composure under pressure, allowing them to control the tempo of a counterattack effectively. When defending high or in advanced positions, their involvement in the transition phase enables a swift shift from defense to offense, capitalizing on the element of surprise and exploiting defensive imbalances [22]. These positional roles are fundamental to the effective execution of counter-attacking strategies. Coordinated movements across all lines, paired consists of only a few events with swift transitions, are key to maximizing the success of counterattacks.

Besides, the factor advanced distance (delta x), which reflects a player's ability to advance the ball quickly, plays a critical role in facilitating rapid transitions from defense to attack. This ability to progress the ball efficiently can disrupt defensive structures and create dynamic offensive opportunities. In fast-paced games, the quicker a player can move the ball, the more likely they are to exploit defensive gaps before the opposition can reset [46]. A higher delta x value thus correlates with a player's tactical awareness and ability to accelerate play, significantly contributing to a team's overall transition effectiveness.

Equally important is the relative angle to the actor, which plays a pivotal role in shaping a player's decision-making process [47]. The angle and distance provide critical context for the options available to the player in real-time. The angle influences the number of potential options available for passing, shooting, or dribbling. A favorable angle might provide multiple passing lanes or shooting opportunities, whereas an unfavorable angle could restrict these options, leading to a higher likelihood of turnovers or suboptimal passes [48]. This relationship between the angle and the available decision-making options underlines the importance of the player's spatial awareness and adaptability in high-pressure moments. Such finding suggests players with the ability to adjust their body position to gain a more advantageous angle are better equipped to make precise, rapid decisions in transition phases.

Moreover, it could be indicated that delta x and the relative angle are interdependent. A player who can quickly advance the ball with a high delta x may also position themselves at a more favorable angle, thereby enhancing the quality of their decision-making. This synergy between speed and spatial awareness is crucial in fast-paced transitions and emphasizes the need for players to maintain both physical and tactical awareness during the match [49]. In practical terms, coaching strategies should aim to develop not only the player's speed and physicality but also their ability to read the game and adjust their positioning for optimal decision-making.

Interestingly, features such as VAEP, OBV, and goal difference were found less contributive to the model's predictive power but were still relevant to counterattack outcomes compared to player positioning and movement-related metrics. Such phenomenon might stem from the unique nature of counterattack, which relies on speed

and precision [10] While these metrics are useful for assessing the probability of a team scoring and conceding as a direct result of the first ten actions. They may fail to capture the rapid spatial dynamics and quick decision-making needed during counterattacks [21, 50]. In these scenarios, spatial awareness and timing have a much more direct impact on counterattack success, as the key lies in exploiting the opponent's temporary disorganization [50, 51]. Broader metrics that evaluate action value across multiple phases of play may therefore be less effective in these fast and decisive moments. On the other hand, some features, such as event play patterns, show more variability in importance, suggesting their impact may depend on specific situations. This analysis helps identify the most important features of the model, guiding effective feature selection to improve performance [52].

This finding aligns with previous research that highlights player tactical position and movement as key components of successful offensive plays [53], while also extending this understanding by quantifying the relative contributions of each factor to counterattack outcomes. Specifically, while prior studies have emphasized the tactical importance of defenders and midfielders during transitions [54, 55], our findings go further by explicitly quantifying their roles. The analysis shows that defenders, particularly those in positions 3, 5, and 7, play a crucial role in initiating counterattacks, effectively transitioning play into offensive opportunities by capitalizing on depth and exploiting the space behind opponents [10]. Additionally, tactical systems that employ three center-backs, such as the 3–5–2 or 3–4–3 formations, significantly enhance the attacking potential of wide players, particularly right-wingbacks [56]. The width these wingbacks provide is essential for stretching the opposition's defense and creating space in the final third. This tactical formation often results in a higher counterattack conversion rate for both right-wingbacks and center-backs. The increased attacking freedom afforded to these players within a back three setup allows them to position themselves in more advanced, goal-scoring opportunities [57]. This underscores the importance of defensive players not only in preventing goals but also in enabling offensive transitions. The observed variability in contributions from midfielders and forwards suggests a strategic advantage in recovering possession close to the opponent's goal, thereby increasing scoring opportunities. This points to the context-dependent role of midfielders and forwards, highlighting their situational influence during counterattacks and the need for adaptable strategies based on player roles[46]. It indicates defenders' significant role in initiating counterattacks, with midfielders contributing to key passes for shot attempts. It highlights the need for players to enhance their positional skills and tactical awareness to fully capitalize on these opportunities.

It is acknowledged that the current study has several limitations. First of all, the available dataset is incomplete as it neither contains full spatio-temporal information for players nor provides identifiable their identification. The absence of this crucial information limits the model's ability to account for off-camera player interactions, making it challenging to fully capture the dynamics of the entire team during counterattacks. The inability to identify individual players further restricts the analysis,

as player-specific skills, roles, and tendencies, which are key in shaping the outcomes of counterattacking situations, are missing from the evaluation. Consequently, the predictive model may not fully capture the nuanced interplay between all players on the field. Secondly, while our model effectively incorporates features like player tactical position, event type, and timing, other influential aspects remain unaddressed. Specifically, physical indicators such as player fatigue, speed, and stamina are critical in counterattacking situations but are not captured by our dataset [58, 59]. These physical factors directly influence players' ability to sustain high-intensity runs, execute precision tactical intentions, and maintain performance under pressure. Similarly, psychological factors, which affect decision-making, confidence, and overall mental resilience, are not considered [60]. As a result, these omitted factors may lead to an incomplete understanding of the conditions contributing to successful counterattacks. However, these findings highlight the complexity of offensive tactics in football, where successful counterattacks depend on more than just frequency. Teams focusing on counterattacking should consider not only the quantity but also the context, quality, and execution of these plays to maximize scoring opportunities.

In future work, it is essential to address these limitations by expanding the dataset to include complete player tracking data that provides detailed positioning of all players on the field, as well as their identities. This would allow for a more thorough analysis of individual roles and interactions during counterattacks. Furthermore, the incorporation of physical metrics such as fatigue levels, speed, and stamina could significantly enhance the model's predictive accuracy regarding counterattack outcomes. Simultaneously, the development of a framework that offers multiple offensive options in similar scenarios would further increase the model's adaptability, thereby enhancing its practical application in coaching. This could improve in-game decisions and help coaches optimize player actions during critical counterattacks.

Conclusion

This study effectively extracts counterattack sequences and examines the key metrics that influence counterattack decisions. By breaking down the decision-making process, it introduces an innovative approach to analyzing football strategies by integrating Graph Attention Networks (GATs) with Transformer architectures, enhancing both predictive accuracy and strategic decision-making during counterattacks. This novel combination not only facilitates the prediction of optimal decisions during crucial game moments, but also deepens our understanding of the dynamics that influence these decisions, as well as features that contribute to successful decision-making during these fast-paced offensive moments. The current results allow sports professionals to identify both common and distinct insights from the model, helping them evaluate actions and decision-making effectively. Specifically, the findings challenge the belief that more counterattacks boost goal scoring, emphasizing instead the importance of quality, timing, and players' positional skills. The insights generated by our framework enable

football players and coaches to analyze behaviors in historical games, demonstrate how to devise optimal offensive strategies to counteract specific team defenses, and assess the likely outcomes of tactical decisions before implementation. This capability provides a substantial advantage in understanding optimal counterattack strategies and making informed tactical choices that adapt to dynamic match conditions.

Supplementary Information

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Supplementary Material 1

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Author contributions
YJY conceptualized the studies and wrote the initial manuscript. CYX and GHM validated the revised manuscript.

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Availability of data and materials
No datasets were generated or analysed during the current study.

Declarations

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Not applicable.

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The authors declare no competing interests.

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